

High-speed CAN Baud rates ranging from 5 Kbit/s to 1 Mbit/s are available, depending on the cable length. This standard is widely used for the physical layer due to its simplicity in connecting devices with cables. It is also the physical standard in the DeviceNet and CANopen specifications. High-speed CAN networks require a 120 ohm resistor to be terminated at each network endpoint.

Low-Speed/Fault-Tolerant CAN Baud rates ranging from 5 to 125 Kbit/s are available. The CAN bus communication can continue even in the event of a connection failure. In low-speed/fault-tolerant CAN networks, each device has its own terminal.

Note: Due to the different types of terminals, it is not possible to use high-speed and low-speed/fault-tolerant CAN devices in the same network.

The CAN bus is divided into three types: single-wire CAN, low-speed fault-tolerant CAN, and high-speed CAN. Before discussing their differences, it is important to clarify the following points:

1. The classification of buses as Class A, B, or C is irrelevant in the context of high-speed CAN, as it can operate at the same baud rate as low-speed CAN. However, if low-speed CAN uses the baud rate of high-speed CAN, a problem arises. This issue is not solely caused by the baud rate, but also by the transceiver. The reason for this is that the transceiver for low-speed CAN has a lower level conversion efficiency compared to the transceiver for high-speed CAN.

2. There is no difference between them in terms of bit transmission timing, bit arbitration, errors, checksums, and frame structure at the data link layer. The protocols above the data link layer, such as the transport layer, network layer, session layer, and application layer, are artificially specified.

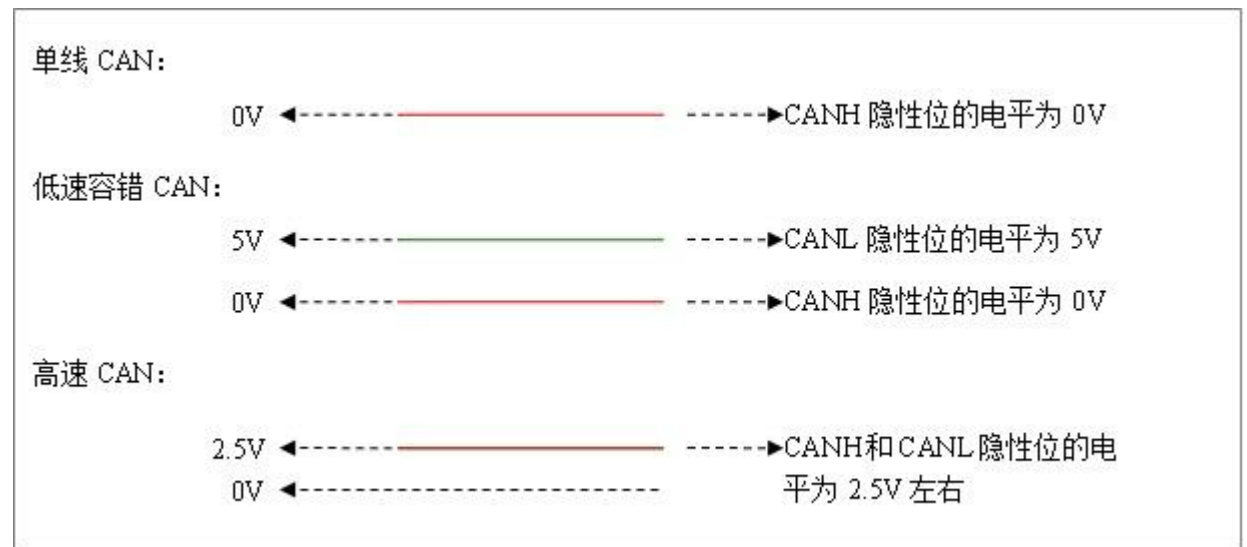
3. The physical differences between these three buses are primarily related to their transmission medium and number of wires. Single-wire CAN utilises only one

wire, while low-speed fault-tolerant CAN and high-speed CAN are transmitted over twisted pairs.

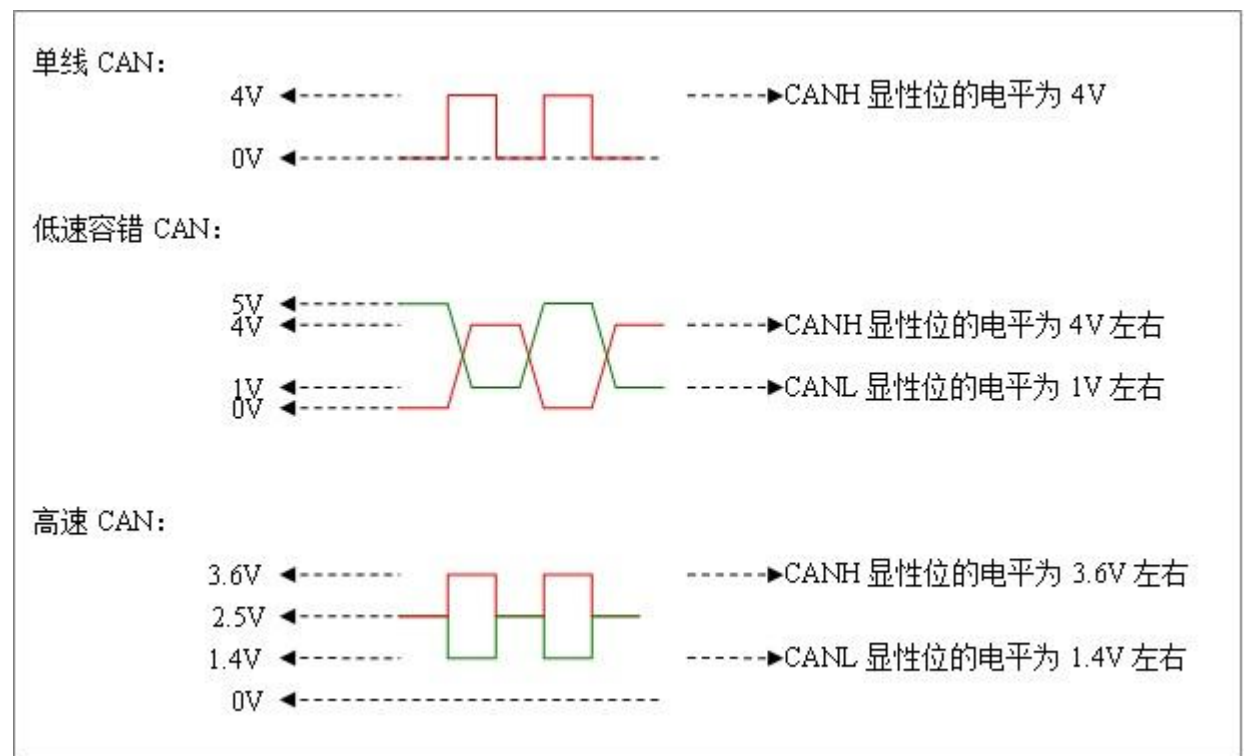
4. The main difference between these three buses is their level of logic.

This can be illustrated through their idle time, active time, sleeping time, and working time. It is assumed that the transceiver is supplied with standard 5V and 12V.

1. Idle (Implicit Bit)



2. Effective (Dominant Bit)



3. Sleep

单线 CAN:

0V ←-----→ CANH 睡眠时的电平为 0V

低速容错 CAN:

12V ←-----→ CANL 睡眠时的电平为 12V

5V ←-----→

0V ←-----→ CANH 睡眠时的电平为 0V

高速 CAN (收发器不同表现不一样):

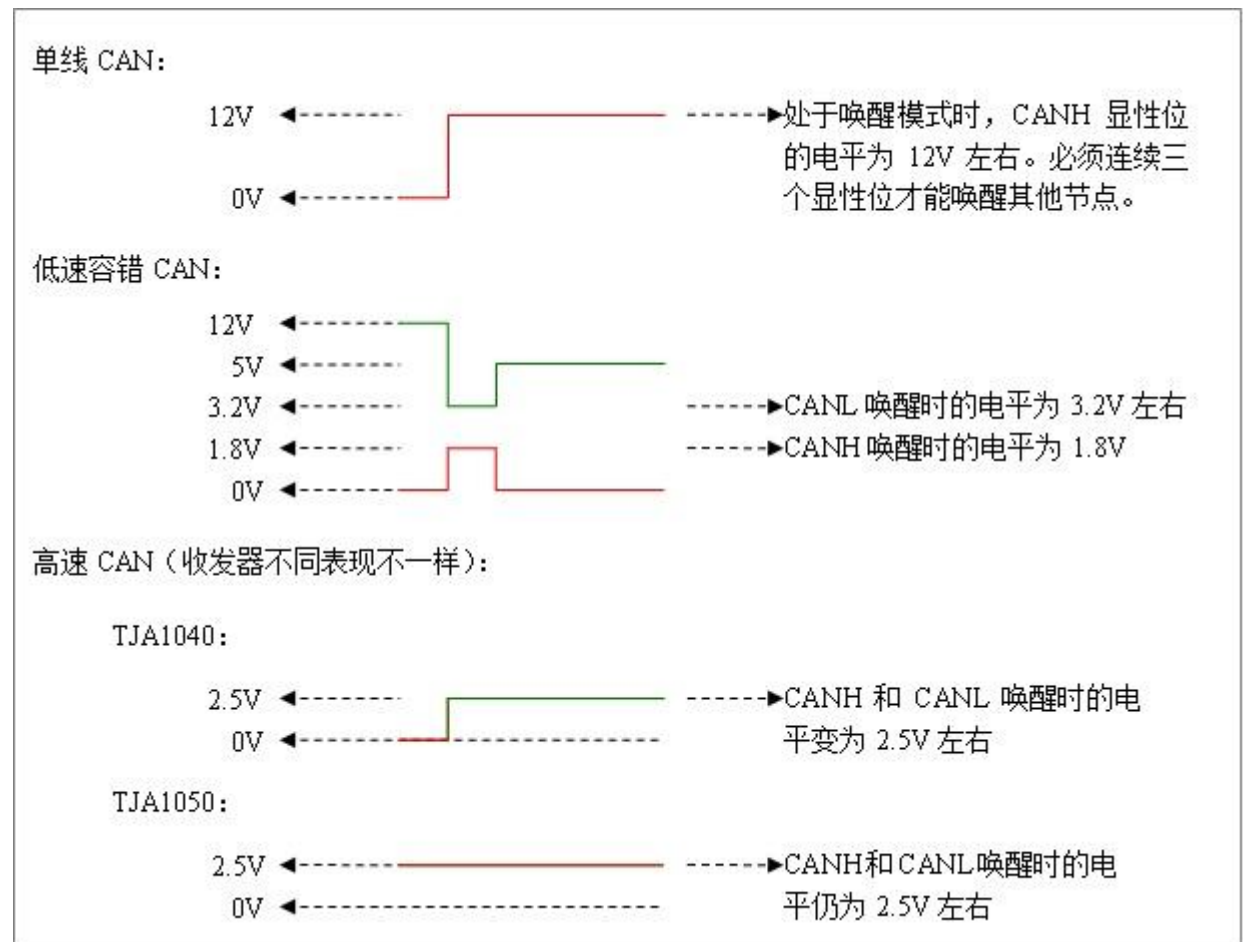
TJA1040:

2.5V ←-----→
0V ←-----→ CANH 和 CANL 睡眠时的电平为 0V

TJA1050:

2.5V ←-----→ CANH 和 CANL 睡眠时的电
0V ←-----→ 平仍为 2.5V 左右

4. Working



These physical characteristics of the CAN bus are directly reflected in the transceiver. No matter which CAN bus, the interface with the microcontroller (TXD and RXD pins) is the same after passing through the transceiver, and the level logic at this point will also become consistent.